Automating 2D Classification in Cryo-Em Single-Particle Reconstruction Pipeline

Bhrij Patel, Jeevan Tewari

E-mail: bhrij.patel@duke.edu, jeevan.tewari@duke.edu

April 2021

Abstract.
In Cryo-EM, identifying which particles to use for 3D reconstruction and which to discard is a crucial component of the single-particle reconstruction process. One traditional used method to navigate the noisy data and develop a usable particle stack is 2D classification. This process often requires human input in differentiating between good and bad classes. In this report, we provide two deep learning approaches that identify a distinction between good and bad particles to automate this process. We provide both supervised and unsupervised approaches and examine their accuracies across three datasets. In all three datasets, the supervised method outperforms the unsupervised.

1. Introduction

In cryo-electron microscopy (Cryo-EM), an electron beam is used to image vitrified biological samples. In producing these images, protein particles that are randomly oriented in 3D space are scanned, producing micrographs. These resulting images contain 2D projections of the protein from many views and these different projections can be utilized to obtain a 3D electron density map in single-particle reconstruction (SPR).

Electron doses are limited in this imaging pipeline to reduce radiation damage to the biological material. A byproduct of this low dose is a low signal-to-noise ratio (SNR). This low SNR is exacerbated by convolution with the contrast transfer function (CTF) which is performed during measurement by the electron microscope. One established method to ameliorate this issue of low SNR is to align patches of similar orientation and take the average of this alignment. This process addresses the loss of data from applying the CTF by combining images from different defocuses. Visualizations of these averages can aid in determining what particles to use for 3D reconstruction and which particles to discard.

To identify particles with similar alignment, a 2D classification algorithm can be used. This algorithm involves iteratively performing k-Means clustering and class

averaging. This strategy rests on the assumption that there is a finite set of views that the particles occupy.

After several iterations, the class averages can be evaluated to determine which particles should be used in the particle stack for further processing. Additionally, these results provide visual interpretations of the data and allow researchers to identify particular aspects of the protein, in some cases even secondary structures are visible in these averages [5]. In this visual analysis, classes are often manually selected to generate a usable particle stack.

Attempts have been made to incorporate 2D class averaging into particle picking to decrease false-positive selection in this step [3]. The researchers behind the Cryo-EM software package RELION have produced a maximum likelihood approach to 2D classification that is currently widely used [4]. Other improvements to this step include the construction of Pre-pro, a fast pre-processing method that uses a dimension-reduction denoising strategy [1]. It is the purpose of this study to develop a model that can perform this task without manual input.

2. Data Collection

2.1. General Data Collection Methods

Three datasets were used for this study: the Apoferritin tutorial dataset for cisTEM (EMPIAR-10146), β-Galactosidase (EMPIAR-10017), hemichannel INX-6 (EMPIAR-10291). Documented resolutions for these entries are 3 Å, 3.2 Å, and 3.8 Å respectively. For the Apoferritin dataset, unaligned movie frames were imported into a cisTEM environment hosted on a virtual software container [2]. Micrograph scans were downloaded from the β-Galactosidase and hemichannel entries and similarly imported into cisTEM – this formatting was preferred as micrographs conserve significant computational space. Particle picking was performed within cisTEM to produce the input patches for 2D classification.

2D classification was performed with varying numbers of classes per dataset. It is recommended that no less than 200 images are contained in each class to ensure reasonable class averages are produced [2]. Approximately 2000 images were generated from the particle picking of Apoferritin so 10 classes were used whereas >20,000 images were produced for β-Galactosidase so 20 classes were used. 40 classes were used for the hemichannel as there were >60000 images. Particles contributing to the classes that looked acceptable to use for 3D reconstruction were added to one particle stack while the discarded particles to another. This process creates what we refer to as the "good" and "bad" particle stacks. Examples of particles from the stacks used for 3D reconstruction are shown in 1. This partition between used and discarded particles represents the main partition used to train and test the models.

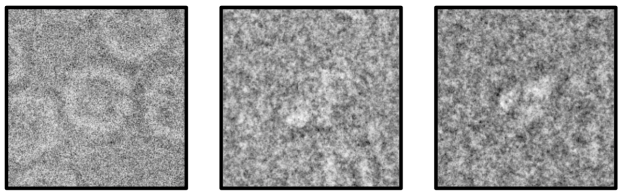

Figure 1. Example particles from the stacks used for reconstruction. From left to right they are Apoferritin, β-Galactosidase, and hemichannel INX-6.

	+	-
Predicted +	1428	0
Predicted -	0	1463

Table 1. Confusion Matrix of the Apoferritin Training Set during test time. We see that the discriminator learned to 100% correctly classify the particles.

2.2. Addressing Imbalances in the Data

The datasets generated each had a heavy imbalance in labels with the bad stacks having much fewer particles relative to the good stacks. For example, the Apoferritin dataset had about 2000 good particles and fewer than 60 bad particles. In order to remedy this skew, the bad particles were randomly rotated and copied until the particle stacks were approximately even.

3. Methods and Results

3.1. Binary Classifier

The first method is a supervised learning approach. In this approach, we train a basic discriminator or binary classifier that takes in a particle image during training and outputs the probability of that particle being a good particle. We then compared the output probability to the label of the image. Labels are assigned the value of 1.0 if they a represent good particle and 0 otherwise. We take the probability and the label and perform binary cross entropy loss. Let p_i be the probability output for input image i and let y_i be the corresponding label. This loss function is displayed in equation 1.

$$\sum_{i}^{n} y_i \log p_i + (1 - y_i) \log(1 - p_i) \tag{1}$$

We trained three models, one for each of the datasets. From Figure 3, we see that when training a binary classifier on the data, the hemichannel dataset had a much higher loss initially and then converges to the same loss as Apoferritin and β-Galactosidase. The losses seem to converge at 80 epochs, signaling that the discriminator has found an optimal solution.

For Apoferritin, the discriminator the on training set is able to classify the particles with 100% accuracy and 98% accuracy on the test set (Table 1, 2). This little drop in

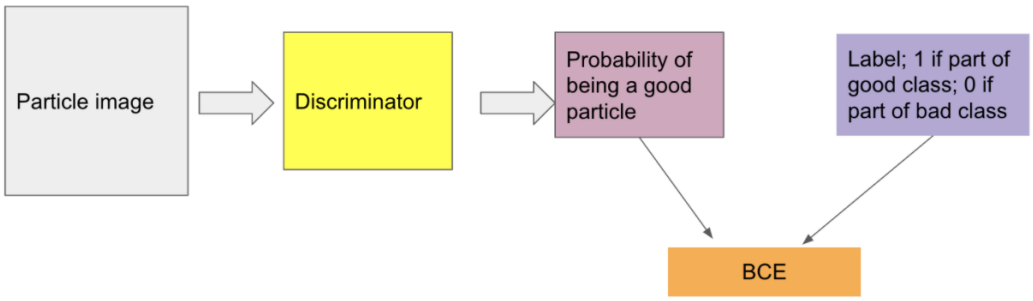

Figure 2. Binary classifcation: particle image 64×64 is inputted into the network and a scalar value between 0 and 1 is outputted. This is taken as the probability that the particle image is that of a good particle. Binary cross entropy loss is then calculated with this probability and the label.

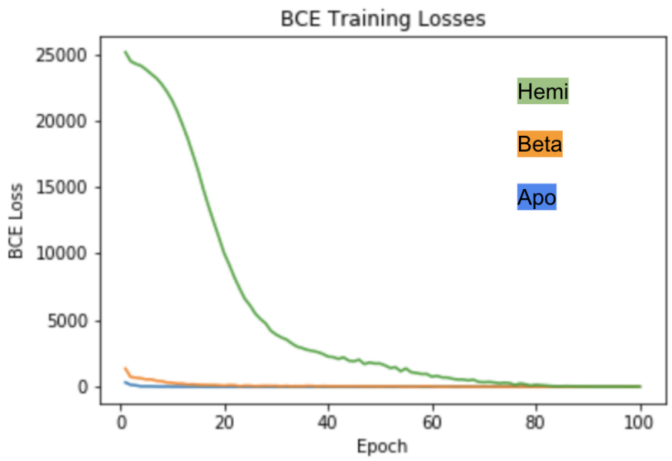

Figure 3. BCE loss of the three datasets during training of this discriminator. hemichannel has a significant difference between the other datasets in the beginning but then converges to the same loss after about 80 epochs.

	+	-
Predicted +	627	20
Predicted -	0	632

Table 2. Confusion Matrix of the Apoferritin training set during test time. We achieve 98% accuracy, suggesting the model did not overfit the training data.

accuracy suggests that the model is generalized well to classify new Apoferritin images it has not seen yet.

For β-Galactosidase, the discriminator, for both the training set and test set, is able to classify the particles with 97% accuracy (Table 3, 4) on the test set. So while accuracy decreased from Apoferritin, the model is more resilient to overfitting.

The model trained on hemichannel images had the worst generalization and there is clearly overfitting. The accuracy on training set is 99% but 73% on the testing, (Table 5, 6). We believe, like for the high BCE loss, this is due to the quality of hemichannel

	+	-
Predicted +	18930	890
Predicted -	1	18560

Table 3. Confusion Matrix of the β-Galactosidase training set during test time. The discriminator learned to 97% correctly classify the particles

	+	-
Predicted +	8013	477
Predicted -	33	7763

Table 4. Confusion Matrix of the β-Galactosidase training set during test time. As with the training set, we achieve 97% accuracy, suggesting almost no overfitting.

	+	-
Predicted +	25343	201
Predicted -	92	24364

Table 5. Confusion Matrix of the hemichannel training set during test time. The model achieves 99% accuracy with the training set.

	+	-
Predicted +	9167	3508
Predicted -	2702	7968

Table 6. Confusion Matrix of the hemichannel Testing Set during Test Time. Overfitting is apparent from the drop to 73% accuracy.

images.

3.2. Autoencoder and K-Means Clustering

While the first approach performed well for two out of three of the datasets, it still relies on labels that were generated from 2D classification from cisTEM, which does not entirely remove the need for manual 2D classification selection. We now present our unsupervised approach.

We attempted to combine an autoencoder and K-Means clustering to find distinctions between good and bad particles in some latent space. Using an autoencoder, we pass in the particle images as the input and then try to get back perfect reconstruction using L_1 reconstruction loss. Let x_i be input image i and let $\hat{x}_i$ be the reconstruction. The loss function can be found in equation 2.

$$\sum_i ||x_i - \hat{x}_i||_1 \tag{2}$$

If we can learn a faithful reconstruction using this model then autoencoder has also

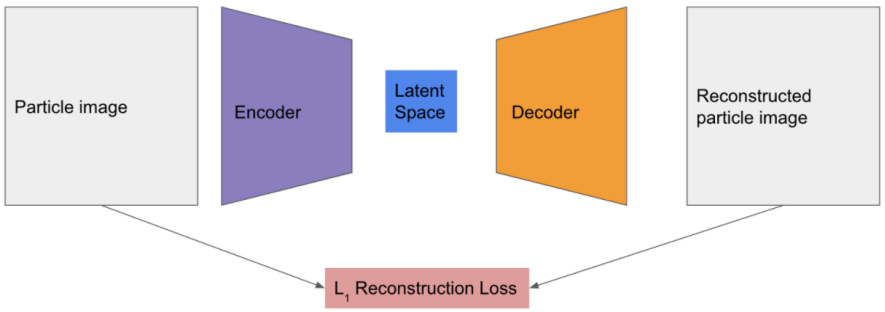

Figure 4. Autoencoder diagram. The particle image is fed inside the encoder and the output is a latent vector that is then sent to the decoder to reconstruct back to the original dimension (64×64. Taking the original input and the reconstructed output, we do L_1 reconstruction as the training loss to update the entire network.

hopefully learned a faithful representation in the latent space that be used for clustering.

From Figure 5 we see a wider disparity in losses between the three datasets. While all losses seem to have converged by the 100-th epoch, the hemichannel dataset is much greater than the β-Galactosidase which is much larger than Apoferritin. We believe this may be because the quality of images of hemichannel is significantly worse than Aporferritin and β-Galactosidase.

After training the autoencoder, we pass in all the training images during test time. The set of latent vectors representing each image is outputted by the encoder and are then clustered by using K-Means to form k clusters. After obtaining the clusters, we then selected certain clusters to be chosen as good clusters. Similar to 2D classification, K-Means clusterings is also an iterative process that changes the averages of clusters until some convergence or max number of steps. During test time, a particle is inputted into the encoder. If the latent vector outputted is part of a good cluster, it will be classified as good; otherwise, it is classified as bad. The driving motivation of training an autoencoder to create a faithful latent space and clustering in that space rather than just clustering in the original image space is that K-Means clustering generally performs better in lower dimensions.

Two questions arise in this set-up: How many clusters do we form in the latent space and which clusters do we select? The first question foundational to the K-Means algorithm. For the number of k clusters, we simply chose $k = 10, 20, 40$. Since the latent space we cluster stays the same between different values of k, we do not need to retrain the models, just run K-Means again. For the selection of clusters, we chose the m largest clusters to be classified as good with the theory that the bigger the cluster, the better the particles in them. For each value of k we ran K-Means with, we tried all possible of m. Meaning for $k = 20$, we first only chose the largest cluster to be a good cluster and see how well the classification was. We repeated this, incrementing m by one each time until we got to k.

In this report we present, for each dataset, the performance of the best k and m values on the training set and their performance on the testing set. So for Apoferritin,

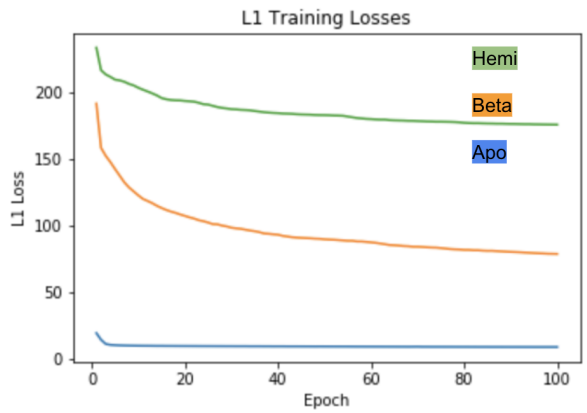

Figure 5. L1 Loss for Autoencoder for Apoferritin, β-Galactosidase, and hemichannel. There is a significant difference in the losses and they converge at distant values.

	+	-
Predicted +	1428	79
Predicted -	0	1384

Table 7. Confusion Matrix, $k = 10, m = 3$, of the Apoferritin training set during test time. We see that the auotencoder and K-Means learned to 97% correctly classify the particles.

	+	-
Predicted +	627	53
Predicted -	0	599

Table 8. Confusion Matrix, $k = 10, m = 3$, of the Apoferritin training set during test time. With 96% accuracy, the model did not overfit the training data.

	+	-
Predicted +	4986	602
Predicted -	13945	18848

Table 9. Confusion Matrix, $k = 20, m = 1$, of the β-Galactosidase training set during test time. This setup achieved 62%. We see that there are nearly 14000 False Negatives, suggesting that many of good particles ended up in clusters smaller than the largest one.

choosing the 3 largest clusters out of ten did the best in training, so we also present that setup‘s performance on the test set.

The model achieves great performance with Apoferritin with 97% training and 96% (Table 7, 8). The only misclassification were False Positives, meaning that some negatives ended up in the 3 biggest clusters.

With β-Galactosidase, we see a drop in performance in training set with 62% accuracy (Table 9) and then 51% with the test set (Table 10). The model does not generalize well and the clusters formed on the training set do not separate good from

	+	-
Predicted +	8222	18
Predicted -	8045	1

Table 10. Confusion Matrix, $k = 20, m = 1$, of the β-Galactosidase testing set during test time. This setup achieved 51% on the test set. So not only did the model not achieve desirable performance on the training set, it also did not generalize to unseen images.

	+	-
Predicted +	19215	9416
Predicted -	6220	15149

Table 11. Confusion Matrix, $k = 20, m = 5$, of the hemichannel Training Set during Test Time. This setup achieved 69%

	+	-
Predicted +	5445	6031
Predicted -	0	11869

Table 12. Confusion Matrix, $k = 20, m = 5$, of the hemichannel test set during test time. It surprisingly achieved 74%, doing 5% better than the training set.

bad particles and do not generalize well on other images.

Surprisingly, on the hemichannel dataset, the test set outperformed the training set by 5% (Table 11, 12). hemichannel even outperforms β-Galactosidase even those the L_1 loss is higher (Table 5).

4. Conclusion and Next Steps

When comparing the accuracies of these two approaches, the supervised method outperforms the unsupervised approach, especially with β-Galactosidase and hemichannel. From these results, we see that utilizing labels significantly helps in creating distinctions between good and bad particles. We also believe that image quality plays a factor as the INX6 and β-Galactosidase both performed worse than Apoferritin.

An experiment we would like to try in the future would be to select clusters based on variance rather than size. With this modification, the pipeline will be able to accurately discriminate even when the input dataset has more bad particles than good. The current pipeline would identify the largest group of bad particles, classify them as good, giving an egregious amount of false positives.

5. References

[1] Szu-Chi Chung, Hsin-Hung Lin, Po-Yao Niu, Shih-Hsin Huang, I-Ping Tu, and Wei-Hau Chang. Pre-pro is a fast pre-processor for single-particle cryo-em by enhancing 2d classification. *Communications Biology*, 3(1):508, Sep 2020.

[2] Timothy Grant, Alexis Rohou, and Nikolaus Grigorieff. *cis*tem, user-friendly software for single-particle image processing. *eLife*, 7:e35383, mar 2018.

[3] Donal McSweeney, Sean M. McSweeney, and Qun Liu. A self-supervised workflow for particle picking in cryo-EM. *IUCrJ*, 7(4):719–727, Jul 2020.

[4] Sjors H. W. Scheres. Relion: implementation of a bayesian approach to cryo-em structure determination. *Journal of structural biology*, 180(3):519–530, Dec 2012. 23000701[pmid].

[5] Fred Sigworth. Principles of cryo-em single-particle image processing. *Microscopy*, 65(1):57–67, 2015.